Rational Number Project

Fraction Operations and Initial Decimal Ideas **Lesson 28: Overview**	**Materials**
Students build on their experiences with pictures, contexts, and benchmarks of $\frac{1}{2}$ and 1 to estimate quotients to fraction division problems.	• Student Pages A, B, and C • Overhead of student pages for teacher • Overhead of Student Page C for teacher (optional for students)

Teaching Actions

Warm Up

Imagine drawing a picture to solve this problem. From that picture in your mind, find the actual answer.

$3\frac{1}{2} \div \frac{1}{4}$

Large Group Introduction

1. Begin the class with these estimations problems. Students should have a copy of Student Pages A and B with these problems.

2. Ask students to imagine the picture they would draw to solve the first problem.

   You have 4 cups of flour. The recipe you are making calls for $\frac{2}{3}$ cup of flour. About how many full recipes can you make?

   Ask: If the recipe called for $\frac{1}{2}$ cup of flour, how many recipes can you make? But the recipe calls for $\frac{2}{3}$ cup of flour; will you have more or less than 8 recipes? Do you think you will have more than 4 recipes? Explain your ideas.

3. Repeat for problem 2 on Student Page A.

Comments

Encourage students to explain their estimates based on order, equivalence ideas, and mental images students have for each problem.

For example, to estimate $4 \div \frac{2}{3}$, a student could reason that the answer must be greater than 4 because $\frac{2}{3} < 1$. The answer will be less than 8 because $\frac{2}{3} > \frac{1}{2}$. If it was $4 \div \frac{1}{2}$, the answer is clearly 8.

Using 1 and $\frac{1}{2}$ as benchmarks will help them in their estimation.

Teaching Actions

You have $2\frac{1}{2}$ yards of ribbon. You cut it into pieces $\frac{3}{4}$ in length? Estimate: About how many full pieces can you cut? At least 2? At least 5?

Guiding Questions: If you cut a length of 1 yard, how many pieces? If you cut a length ½ of a yard, how many pieces? $\frac{3}{4}$ is greater than ½. Is the total number of pieces more or less than 5?

4. Repeat for problem 3 on Student Page A

Estimate by finding the whole number closest to the exact answer: $3\frac{2}{3} \div \frac{1}{4}$.

Guiding Questions*:* How many fourths in three? Is $\frac{1}{4} >$ or $< \frac{2}{3}$? Can you get two more fourths out of the $\frac{2}{3}$?

Small Group/Partner Work

5. Assign the rest of Student Pages A and 2B to students to work with a partner. Guiding questions embedded in the problems lead students to their estimates.

Wrap Up

6. Use Teacher Page C to guide students to find the exact answer to three problems from the work completed in this lesson.

7. Ask for their estimate first. Then ask them to imagine the picture they would draw to solve the problem. Ask: How would they partition that picture to make solving the problem easy?

Ask: What is the number sentence with common denominators for this problem? Find the exact answer. Possible estimation strategies follows:

Comments

Teaching Actions	Comments
a. $2 \div \frac{2}{5}$. If the problem was $2 \div \frac{1}{2}$ then the answer would be 4. $\frac{2}{5} < \frac{1}{2}$ so the answer will be more than 4 but not much more as $\frac{2}{5}$ is close to $\frac{1}{2}$. ($\frac{10}{5} \div \frac{2}{5} = 5$).	
b. $3\frac{1}{2} \div \frac{3}{4}$. If the problem was $3\frac{1}{2} \div \frac{1}{2}$ the answer would be 7. If it was $3\frac{1}{2} \div 1$, the answer would be $3\frac{1}{2}$. $\frac{3}{4}$ is greater than $\frac{1}{2}$ but less than one so the answer is between 3 and 7. ($\frac{14}{4} \div \frac{3}{4} = 4\ \frac{2}{3}$)	
c. $3\frac{1}{2} \div \frac{1}{8}$. It would be greater than 24 because there are 24-eighths in 3. But it is less than 32 because there are 32-eighths in 4. ($\frac{28}{8} \div \frac{1}{8} = 28$).	

Translations:

- Story problem to picture to verbal
- Symbol to symbol to verbal
- Symbol to symbol to picture

Imagine drawing a picture to solve this problem. From that picture in your mind, find the actual answer.

$$3\frac{1}{2} \div \frac{1}{4}$$

Division Estimation Problems

(1) You have 4 cups of flour. The recipe you are making calls for $\frac{2}{3}$ cup of flour. Estimate: About how many full recipes can you make? At least 4? At least 8?

(2) You have $2\frac{1}{2}$ yards of ribbon. You cut it into pieces $\frac{3}{4}$ in length? Estimate: About how many full pieces can you cut? At least 2? At least 5?

(3) Estimate by finding the whole number closest to the exact answer:

$$3\frac{2}{3} \div \frac{1}{4} =$$

(4) You know that $2 \div \frac{1}{2} = 4$. About how much would $2 \div \frac{2}{5}$ be? Is it more or less than 4?

(5) You know that $3 \div \frac{1}{2} = 6$. About how much would $3 \div \frac{2}{3}$ be? Is it more or less than 6?

(6) You know that $4 \div \frac{1}{2} = 8$. About how much would $4 \div \frac{3}{4}$ be? Is it more or less than 8?

(7) You know that $2 \div \frac{1}{2} = 4$. About how much would $2 \div \frac{1}{3}$ be? Is it more or less than 8?

(8) You know that $3\frac{1}{2} \div \frac{1}{2} = 7$. About how much would $3\frac{1}{2} \div \frac{3}{4}$ be? Is it more or less than 7?

(9) You know that $3\frac{3}{4} \div \frac{3}{4} = 7$. About how much would $3\frac{3}{4} \div \frac{1}{8}$ be? Is it more or less than 7

Name_____ Lesson 28/Student Page C

Problem	Estimate	Number sentence with common denominators	Picture to verify answer
$2 \div \frac{2}{5}$			
$3\frac{1}{2} \div \frac{3}{4}$			
$3\frac{1}{2} \div \frac{1}{8}$			

Post Lesson Reflection

Lesson_____

1) Number of class periods allocated to this lesson: _____

2) Student Pages used: _____

3) Adaptations made to lesson: (For example: added extra examples, eliminated certain problems, changed fractions used)

4) Adaptations made on Student Pages:

5) To improve the lesson I suggest: